

LSTM Model Fine-Tuning & Hyperparameter Exploration-25/10/25(Friday):

Data Preparation and Preprocessing

The model was trained on the raw, lowercase text of **Shakespeare's *Hamlet***, totaling **162,881 characters**.

- **Vocabulary:** The unique set of characters was identified, resulting in a **vocabulary size of 44** (including letters, punctuation, and whitespace).
- **Sequence Creation:** The text was broken down into training examples using a **sliding window** approach.
 - **Sequence Length:** Each input sequence (\$X\$) contained **100 characters** of context.
 - **Step Size:** Sequences were generated by skipping 3 characters (**STEP=3**).
 - **Data Size:** This process created **54,261 total sequences**.
- **Vectorization:** Both the input sequences (\$X\$) and the target next character (\$y\$) were **one-hot encoded** to prepare them for the neural network. The final input shape was \$(54261, 100, 44)\$ and the output shape was \$(54261, 44)\$.

2. Model Architecture and Training Configuration

The final model used a revised, higher-capacity architecture and fine-tuned hyperparameters.

Component/Hyperparameter	Value	Description			
Model Type	Sequential	A simple stack of layers.			
LSTM Units	512	Increased capacity from an apparent initial attempt to 512 for better learning.			
Input Shape	(100,44)	(Sequence Length, Vocabulary Size)			
Regularization	Dropout(0.2)	A Dropout layer was added before the final Dense layer.			
Output Layer	Dense (44, Softmax)	Outputs a probability distribution over the 44 possible next characters.			
Optimizer	Adam (Learning Rate = 0.0005)	A low learning rate was selected for potentially better convergence.			

Loss Function	Categorical Crossentropy	Appropriate for multi-class classification (predicting one of 44 characters).							
Batch Size	128	Used for training.							
Early Stopping	monitor='loss', patience=5	Intended to stop training if the training loss did not improve for 5 epochs.							

3. Training Results and Analysis

The model was trained for a maximum of 40 epochs with a 5% validation split

Metric	Best Value (Epoch 12)	Final Value (Epoch 40)	Observations																
Training Loss	1.6722	0.1219	Significant decrease in training loss over 40 epochs, indicating the model learned to fit the training data very well.																
Validation Loss	1.7943	3.4368	The lowest validation loss occurred at Epoch 12. After this point, the validation loss began to increase steadily, while training loss continued to fall.																
Training Accuracy	49.55%	97.07%	Very high final training accuracy, confirming a good fit to the training data.																
Validation Accuracy	47.20%	40.97%	The peak validation accuracy was also achieved around the same period as the lowest validation loss (Epoch 12), and it dropped off significantly afterward.																

Conclusion on Training

The model exhibits a clear case of **overfitting** starting around **Epoch 12**. The dramatic drop in training loss (to ≈ 0.12) and corresponding spike in validation loss (to ≈ 3.44) confirms that the model is no longer generalizing well to unseen data after the 12th epoch.

4. Text Generation

A text generation function was defined using a **sampling** technique with a **temperature** hyperparameter (set to **\$0.5\$** for the test run) to control the randomness of the generated text.

- **Seed Text:** "to be or not to be, that is the question"
- **Generated Output:** The output was **incoherent gibberish** ("to be or not to be, that is the questionooa,tytyyrttrmm?imwt yuttrai eyos a lyaaaa;c,irtoeaaaaanwto,ilo eitr t?ll,t,u atac,iilerlsus,cel,eieeryaryaaaaaaatratoarrlyi iar,alditcaiiisl,oeea tttstutwrteeea,e,trinaiacorliuaaa...").

Conclusion on Generation

The poor quality of the generated text, despite the model having its "best weights" (from the EarlyStopping which monitors **training loss**) and having a high training accuracy ($>97\%$), suggests the overfitting issue is severe. The best performing model (at Epoch 12, with the lowest validation loss of 1.7943 and accuracy of 47.20%) was likely *not* restored, as the early stopping monitored *training* loss, which never improved significantly past 40 epochs.

Recommendation

To obtain better text generation, the following steps should be taken:

1. **Correct Early Stopping:** Change the monitor parameter in EarlyStopping from '**loss**' (training loss) to '**val_loss**' (validation loss) to prevent overfitting and ensure the model with the best generalization is restored.
2. **Hyperparameter Tuning:** Explore further hyperparameter tuning, such as increasing the dropout rate or adding more layers (e.g., a second LSTM layer) to improve the model's ability to learn and prevent overfitting.